

OFFICIAL CORPORATE PROFILE

ALTERNATE ENERGIES LIMITED

Pioneering Solar Infrastructure Across the UK & East Africa

Registered Corporate Information:

- Company Registration No: **07705672** (Incorporated 2011)
- Registered Office: 20 Central Avenue, St. Andrews Business Park, Norwich, NR7 0HR, UK
- Primary Operations: London, Norwich & East Africa Regional Network

1. EXECUTIVE CORPORATE OVERVIEW

Alternate Energies Limited – Registered Company 07705672

Incorporated in England and Wales in 2011, **Alternate Energies Limited** (Company Registration No. 07705672) has spent over a decade establishing itself as a specialized solar infrastructure company. Operating from our registered headquarters at St. Andrews Business Park in Norwich, our operations span across **Norwich, London, and East Africa**.

2011

YEAR ESTABLISHED

07705672

COMPANY REG. NO.

3 Regions

LONDON, NORWICH & E.
AFRICA

4 Pillars

CORE BUSINESS UNITS

Strategic Corporate Focus

Alternate Energies Limited specializes in delivering high-yield solar energy solutions through turnkey solar installation, long-term asset maintenance, commercial rooftop partnerships, and direct solar power supply agreements for commercial and industrial clients.

Primary Operating Locations

Norwich Headquarters

Central administration, engineering design, and primary project hub serving the East of England at 20 Central Avenue, St. Andrews Business Park.

London Operations

Commercial rooftop solar development, capital financing structuring, and corporate partner relations desk.

East Africa Division

Direct solar power supply, off-grid industrial microgrids, and commercial rooftop lease deployments in high-irradiance regions.

2. MISSION, VISION & CORE OPERATING PRINCIPLES

Decarbonizing Commercial Power via Solar Infrastructure

Our Corporate Vision

To be the premier solar partner for commercial enterprises and industrial hubs across the UK and East Africa, delivering zero-capital clean electricity and reliable solar infrastructure.

Our Corporate Mission

To lower enterprise energy expenses through high-efficiency solar installations, direct power purchase agreements, and predictive asset maintenance while eliminating industrial carbon footprints.

Guiding Principles Matrix

Core Principle	Operational Execution	Target Outcome
Technical Rigour	Tier-1 PV hardware selection and MCS-compliant installation standards.	Maximizing kWh/kWp system efficiency.
Commercial Integrity	Transparent PPA pricing structure and zero hidden maintenance fees.	Long-term commercial partnerships.
Operational Safety	Strict compliance with UK Health & Safety Executive and regional guidelines.	Zero-incident safety record.
Regional Empowerment	Local engineering workforce development in East Africa and the UK.	Sustainable community value creation.



3. CORE SERVICE PILLARS

Four Primary Business Streams Offered by Alternate Energies Limited

1. Solar Installation

Turnkey engineering, design, and mounting for commercial rooftops, ground arrays, and agricultural facilities.

- Custom structural solar engineering
- High-efficiency Tier-1 PV modules
- Grid interconnect & commissioning

2. Solar Maintenance

Full-lifecycle operational support, preventative servicing, thermographic audits, and performance optimization.

- 24/7 Remote system monitoring
- Inverter repair & replacement
- Panel cleaning & preventative checks

3. Rooftop Partnerships

Lease your unused commercial roof space to Alternate Energies Limited. We install, finance, and operate the solar array.

- Zero upfront capital investment required
- Long-term roof rental income streams
- Immediate property carbon rating upgrade

4. Direct Power Supply

Purchase clean solar energy generated directly from on-site or localized solar assets at guaranteed discounted rates.

- Long-term Power Purchase Agreements (PPA)
- Protection against volatile grid pricing
- 100% renewable power verification

End-to-End Asset Delivery

By controlling the entire value chain—from initial site assessment and installation through long-term maintenance—Alternate Energies Limited eliminates third-party friction and ensures maximum energy yield over a 25-year asset lifespan.

4. DEEP DIVE: SOLAR INSTALLATION SERVICES

Precision Engineering & Turnkey Project Delivery

Alternate Energies Limited delivers bespoke solar photovoltaic installations tailored specifically to commercial, industrial, and agricultural properties. Our installation teams manage every phase from initial structural load testing to high-voltage grid connection.

Installation Workflow & Milestones

Phase	Engineering Deliverable	Quality Standard
01. Site Feasibility	3D LiDAR shading calculation, structural roof load analysis, and electrical capacity survey.	PVSyst Yield Certification
02. System Design	Electrical schematic creation, inverter sizing, string configuration, and cable run routing.	BS 7671 IET Wiring Regulations
03. Procurement	Sourcing Tier-1 mono-PERC/TOPCon modules and industrial string inverters.	IEC 61215 / IEC 61730
04. Construction	Rooftop racking attachment, module mounting, cable containment, and inverter wiring.	MCS Certified Workmanship
05. Commissioning	Insulation resistance testing, string voltage verification, and DNO grid connection handover.	G99 / G100 Compliance (UK)

Commercial Rooftops

Optimized mounting systems for trapezoidal sheet metal, standing seam, and flat membrane roofs without compromising structural integrity.

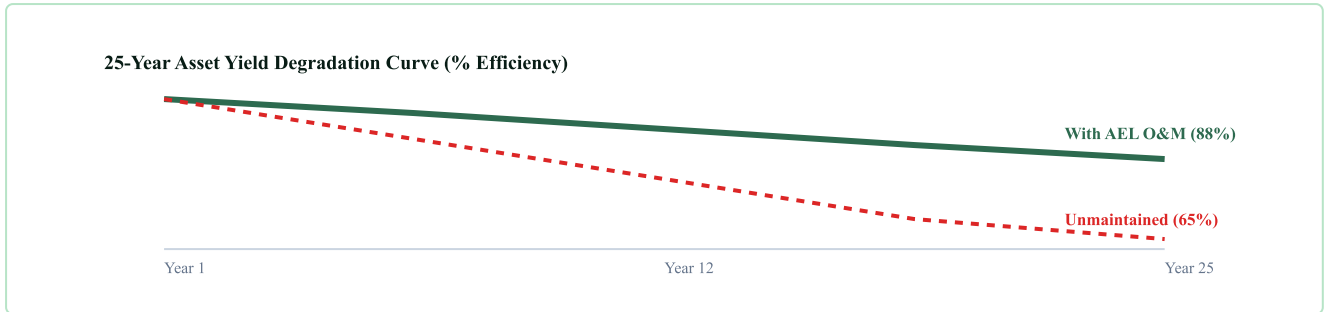
Ground-Mounted Solar

Driven-pile racking systems suited for industrial land, agricultural brownfields, and localized microgrid power generation.

5. DEEP DIVE: SOLAR MAINTENANCE & OPERATIONS

Asset Lifecycle Preservation and Performance Maximization

A solar installation is a 25+ year financial asset. Alternate Energies Limited provides structured maintenance programs engineered to keep operating systems at peak performance and prevent costly inverter down-time.



Key Maintenance Service Offerings

- **Thermal Drone Inspections:** Infrared aerial scans to identify defective cells, hot spots, and string disconnects before catastrophic failure occurs.
- **Automated & Specialized Panel Cleaning:** Removing particulate accumulation, dust, and organic matter in UK and East African environments.
- **Inverter Health Audits:** Replacing worn capacitors, checking cooling fans, and performing firmware updates to maintain 98%+ conversion efficiency.
- **HV & LV Switchgear Testing:** Comprehensive insulation, breaker function, and relay testing to adhere to statutory electrical safety rules.

6. DEEP DIVE: COMMERCIAL ROOFTOP PARTNERSHIPS

Monetizing Unused Roof Space with Zero Capital Expenditure

Through our Commercial Rooftop Partnership model, Alternate Energies Limited rents unused roof space from property owners, industrial warehouse operators, and commercial landlords. We fully finance, install, and maintain a state-of-the-art solar system on the building.

Benefits for Building Owners

- **Guaranteed Rental Income:** Receive indexed rental payments for the roof space over a 20-25 year lease.
- **Discounted On-Site Energy:** Option to purchase electricity generated right on your roof at rates well below grid tariffs.
- **Enhanced Property Value:** Increase building EPC ratings and satisfy tenant ESG requirements.

How the Partnership Works

- **1. Structural Audit:** AEL assesses roof integrity and electrical capacity at no cost to you.
- **2. Lease Agreement:** Simple, standardized roof lease agreement signed between property owner and AEL.
- **3. Zero-CAPEX Build:** AEL funds 100% of installation, grid connection, and ongoing maintenance.

Suitable Property Profiles

Property Type	Minimum Roof Area	Target System Capacity
Logistics Warehouses	1,000 sq. m	150 kWp – 1.5 MWp
Manufacturing Facilities	800 sq. m	100 kWp – 800 kWp
Commercial Retail Parks	1,500 sq. m	200 kWp – 2.0 MWp
Agricultural Buildings	600 sq. m	50 kWp – 300 kWp

7. DEEP DIVE: DIRECT POWER SUPPLY & PPAS

Long-Term Clean Electricity Contracts & Off-Grid Power Solutions

Energy price volatility presents a major risk to commercial operations. Alternate Energies Limited provides ****Direct Solar Power Supply**** agreements (Power Purchase Agreements - PPAs), allowing businesses to lock in fixed, low-cost electricity rates guaranteed for 10 to 25 years.

20-40%
AVG. POWER COST SAVINGS

£0 / \$0
UPFRONT CAPITAL INVESTMENT

10-25 Yrs
PPA CONTRACT TERMS

Comparison: Grid Supply vs. AEL Direct Solar Supply

Feature	Standard Grid Power Supply	AEL Direct Solar Power Supply
Price Stability	Volatile; subject to global fuel shocks.	Fixed or inflation-linked long-term rate.
Carbon Footprint	Contains grid fossil-fuel mix.	100% Zero-Carbon Solar Power.
Transmission Loss	High distribution and network charges.	Minimal (Generated directly on-site).
Capital Investment	N/A	Zero upfront capital required.

East Africa Direct Supply Focus

In East Africa, where central grid reliability can impact business continuity, our Direct Power Supply solutions combine solar PV with battery storage to deliver stable, 24/7 power to industrial and commercial operators.

8. REGIONAL OPERATIONS: LONDON & NORWICH (UK)

Domestic Expertise and Commercial Energy Transition across East Anglia & Capital

Headquartered in Norwich with active project development teams in London, Alternate Energies Limited is deeply rooted in the UK clean energy sector. We support the UK's legally binding target of Net Zero carbon emissions by 2050.

Norwich & East Anglia Operations

Operating out of 20 Central Avenue, St. Andrews Business Park, our Norwich team serves commercial agricultural hubs, food processing facilities, and industrial estates across Norfolk and Suffolk.

- Agricultural barn solar integration
- East of England DNO grid connections (UK Power Networks)
- Local rapid-response maintenance team

London Commercial Hub

Our London development arm manages corporate partnerships, multi-site commercial rooftop portfolios, and structured finance for major real estate investment trusts (REITs).

- Urban logistics rooftop solar arrays
- Corporate PPA structuring
- Tenant energy billing management

UK Regulatory & Technical Compliance

- **Microgeneration Certification Scheme (MCS):** Standard compliance for solar PV installations.
- **Engineering Recommendation G99 / G100:** Full DNO notification and active power control compliance for systems exceeding 16A per phase.
- **IET Wiring Regulations (BS 7671 18th Edition):** Rigorous electrical installation standards guaranteed across every UK project.

9. REGIONAL OPERATIONS: EAST AFRICA FOCUS

Empowering Commercial & Industrial Expansion in High-Irradiance Markets

East Africa represents one of the highest solar irradiance regions globally. Alternate Energies Limited deploys capital and technical expertise to serve commercial, agricultural, and industrial clients across the region.

East Africa Solar Yield Advantage vs. Northern Europe



Key Sector Solutions in East Africa

Agricultural Processing

Solar PV and microgrids for tea, coffee, and horticulture processing facilities to displace diesel generators and lower power overheads.

Manufacturing & Bottling

Direct Power Supply PPAs delivering high-capacity, stable daytime electricity to heavy industrial operations.

Commercial Real Estate

Rooftop solar partnerships for shopping centers, hospitality venues, and logistics centers across primary urban corridors.

Hybrid Microgrid Architecture

For locations with fluctuating central grid voltage, AEL installs smart hybrid systems combining solar PV, lithium battery storage, and automated generator integration to guarantee zero operating downtime.

10. HARDWARE STANDARDS & TECHNICAL SPECIFICATIONS

Tier-1 Hardware Specification and Quality Assurance

Alternate Energies Limited maintains strict component specification guidelines. We exclusively procure from Tier-1 BloombergNEF (BNEF) listed manufacturers to ensure long-term bankability and durability.

Equipment Standard Matrix

Category	Approved Technical Standards	Performance Warranty
Solar PV Modules	N-Type TOPCon / Monocrystalline PERC (550Wp+) Module Efficiency: >21.8%	25-Year Linear Power Warranty (>84.8% at Year 25)
Solar Inverters	Three-Phase Commercial String Inverters Peak Conversion Efficiency: 98.6%	10 to 20-Year Manufacturer Warranty
Mounting Hardware	Anodized Aluminum & Stainless Steel A2/A4 Wind Resistance: Up to 150 km/h	20-Year Structural Integrity Warranty
Battery Storage (BESS)	Lithium Iron Phosphate (LiFePO4) Integrated Thermal Management & Fire Suppression	10-Year / 6,000 Cycle Performance Warranty
Monitoring SCADA	Cloud-based IoT telemetry with module-level or string-level fault alerting.	Real-time continuous uptime SLA

Rigorously Tested for Harsh Climates

All modules and racking components deployed in East Africa undergo salt-mist corrosion, ammonia resistance, and high-dust testing (IEC 61701 / IEC 62716) to withstand severe environmental exposure.

11. REPRESENTATIVE PROJECT PORTFOLIO

Demonstrated Installation and Maintenance Capabilities

Commercial Logistics Hub – Rooftop Solar Array

Norwich, UK

Scope: Turnkey 450 kWp commercial rooftop installation and 25-year O&M agreement under a Rooftop Partnership model.

System Capacity: 450 kWp **Annual Generation:** 425,000 kWh **CO2 Saved:** 95 Tons/Yr

Industrial Processing Facility – Direct Power PPA

East Africa Hub

Scope: 1.2 MWp ground-mounted solar array with integrated 500 kWh battery storage supplying direct clean electricity under a 15-year PPA contract.

System Capacity: 1.2 MWp + BESS **Diesel Displacement:** 65% Reduction **Status:** Operational

Multi-Site Retail Warehouse Portfolio

London & Home Counties

Scope: Long-term solar asset maintenance contract across 12 commercial sites, including thermal imaging audits and scheduled washing.

Portfolio Size: 2.8 MWp Total **Performance Uptime:** 99.4% **Service:** O&M Contract

12. HEALTH, SAFETY & ENVIRONMENTAL GOVERNANCE

Zero-Harm Commitment Across All Operational Sites

At Alternate Energies Limited, working safely on commercial roofs and around high-voltage electricity is our highest priority. We enforce rigorous safety management systems on every project in the UK and East Africa.

HSEQ Operating Pillars

Working at Height Protocol

Mandatory edge protection, fall arrest systems, and certified scaffolding on every rooftop project. All installers hold working-at-height certifications.

Electrical Safety Compliance

Strict Lockout/Tagout (LOTO) procedures during connection and maintenance. Lock-off isolation tested before any work commences.

Environmental Protection

Zero toxic runoff practices during maintenance washing. Recycling programs for all packaging, metal conduit offcuts, and redundant modules.

Quality Management

Structured 40-point quality check before site handover, accompanied by full electrical test documentation and DNO compliance certificates.

Safety Record Statement

Since incorporation in 2011, Alternate Energies Limited has maintained a zero-fatality and zero major lost-time injury (LTI) record across all UK and East African installation and maintenance sites.

13. COMMERCIAL ENGAGEMENT FRAMEWORK

Flexible Financing and Partnership Options for Businesses

Alternate Energies Limited provides flexible commercial arrangements structured around your company's financial preferences and balance sheet objectives.

Commercial Options Overview

Engagement Model	Capital Investment	Ownership Structure	Best Suited For
Direct Outright Purchase	Client Funds 100% CAPEX	Client Owns Asset Immediately	Businesses seeking maximum ROI and tax depreciation advantages.
Power Purchase Agreement (PPA)	Zero CAPEX (AEL Funds 100%)	AEL Owns & Operates Asset	Companies seeking instant energy savings without capital outlay.
Rooftop Lease Partnership	Zero CAPEX (AEL Funds 100%)	AEL Leases Roof Space	Property owners wanting extra rental income from unused roof space.
O&M Service Contract	Annual Service Fee	Client Retains Ownership	Existing solar owners needing professional maintenance.

Free Site & Yield Audit

We provide a fully funded, no-obligation technical assessment and 25-year financial model for eligible commercial buildings. Contact our sales desk at sales@alternateenergieslimited.com to schedule an audit.

14. CORPORATE CONTACT & DIRECTORY

Connect with Alternate Energies Limited

Whether you are looking to install solar on your commercial building, lease your roof space, purchase clean solar power, or partner on regional project development in East Africa, our teams are ready to support you.

Registered Office & Headquarters

Alternate Energies Limited

20 Central Avenue, St. Andrews Business Park

Norwich, Norfolk, NR7 0HR, England, UK

Incorporated in England and Wales | Company Registration No: 07705672 (Est. 2011)

Departmental Contact Emails

Sales & Project Enquiries

sales@alternateenergieslimited.com

Solar installations, roof partnerships, PPA proposals, and site feasibility audits.

General & Corporate Enquiries

info@alternateenergieslimited.com

General corporate information, media relations, and administrative requests.

Purchasing & Supplier Portal

purchase@alternateenergieslimited.com

Tier-1 PV module suppliers, inverter manufacturers, mounting hardware vendors, and regional electrical sub-contractors.

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